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Agenda

Statistics

* Types of statistics
* Mean, Median, Mode
* weighted Average
* Range
* Inter Quartile Range
* Cummulative Distributive Function [CDF]
* Variance, Standard Deviation.

$Q_3 - Q_1$

Simple case study

Cricket

Assignments

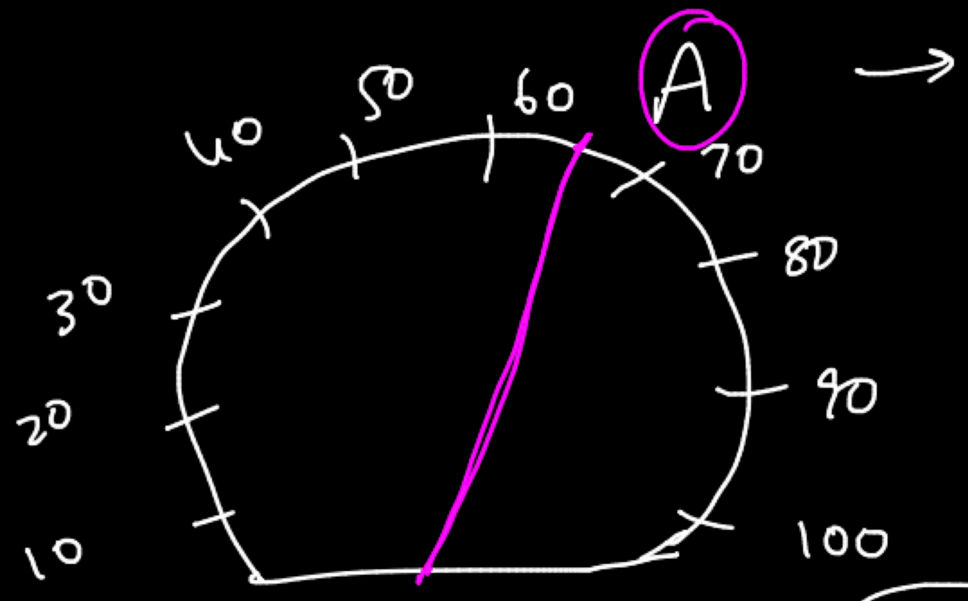
Numpy
pandas

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Types of statistics

(1) Descriptive statistics : Summerizing the data (10)



→ B

65 kmph

5hr

Banglor

Hydrad

Google

Conclusio

100% accu

Inferential statistics

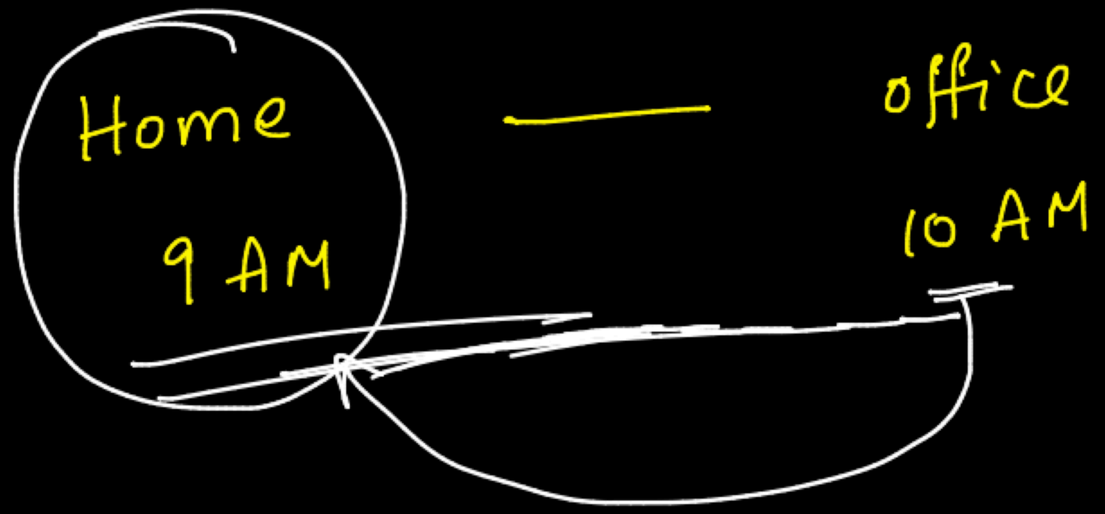
Hypothes

tra

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Krishna



June
1

J30
29

Inferential statistics

⇒ what will they think?

(i) traffic
(ii) accident
(iii) Car got

⇒

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Descriptive : Summerise the data

Influential : Drawing some conclusion based on evidence of data

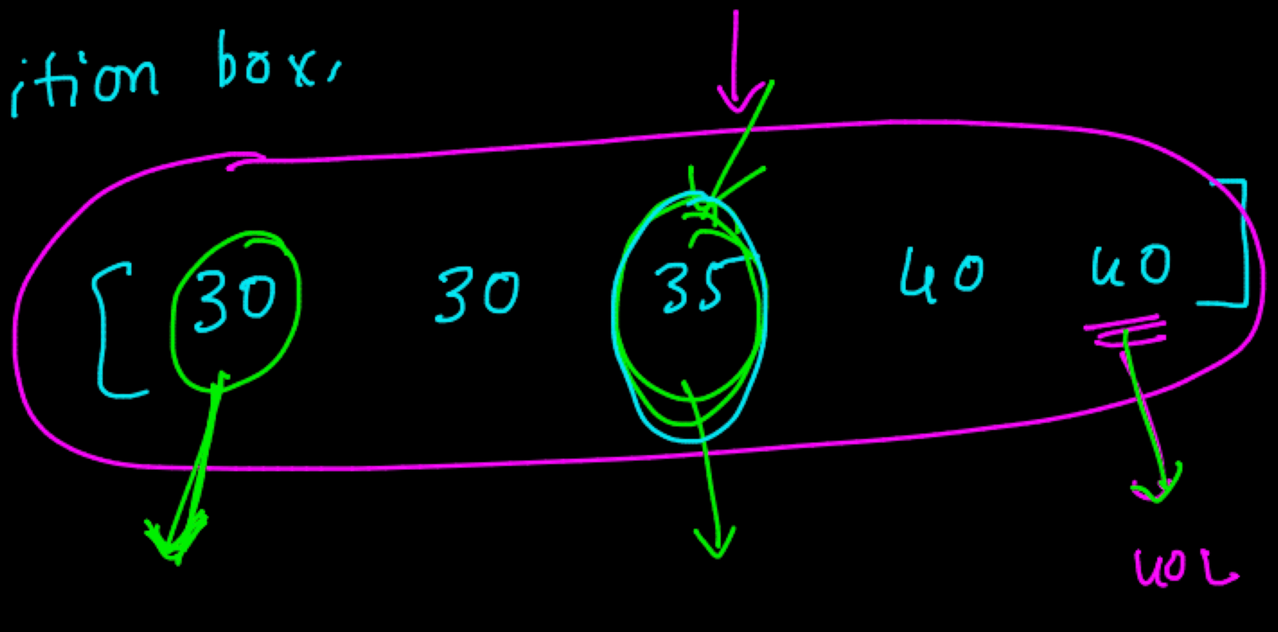
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=> salary range of a data scientist.

glassdoor / ambition box.

Sal - lakhs at google glassdoor



HR

mean | Avg = $\frac{30 + 30 + 35 + 40 + 40}{5}$

= 35 L

median = 35

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⇒

30r

[30 30 35 40 40 300

outlier

Mean = $\frac{30 + 30 + 35 + 40 + 300}{5}$

= 80 L

$\frac{35+40}{2}$ = 37.5

Median is more robust to outliers

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Quiz #1

4 people whose average age is 24

3 people : 20, 22, and 28

(*) median age of these 4 people ?

$$\frac{20 + 22 + 28 + x}{4} = 24 \Rightarrow x = 96 - 70$$

$$x = 26$$

20, 22, 26, 28

$$\frac{22 + 26}{2} \Rightarrow 24$$

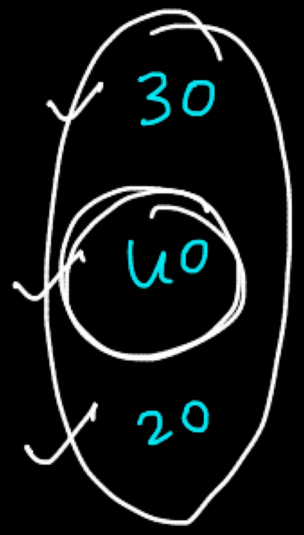
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Mode : The most frequently occuring data point.

90, 85, 90, 90, 91, 87, 90

⇓

90



people said cat ✓

" dog ✓

" cow

mode of favorite animals in this

⇓

dog

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**

30% people had 0 pets
40% people had 1 pet
10% had 2 pets
20% had 3 pets

avg no of pets?

% of people	# of pets
30	0
40	1
10	2
20	3

$$= \frac{30(0) + 40(1) + 10(2) + 20(3)}{100} = \frac{120}{100} = 1.2$$

$$= \frac{(\overbrace{0+0+0+\cdots+0}^{30}) + (1+1+1+\cdots+1) + (2+2+\cdots+2) + (3+3+\cdots+3)}{100}$$

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Ex 2 children in a family is 40 kgs mother is included. mean becomes 45 weight of the mother ?

$$\frac{C_1 + C_2}{2} = 40 \rightarrow (1) \qquad C_1 + C_2 = 80$$

$$\frac{C_1 + C_2 + M}{3} = 45$$

$$80 + m = 135$$

$$m = 135 - 80 = 55$$

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Range => max - min

Salary

[30, 30, 35, 40, 40]

=> 40 — 30 => 10

[30, 30, 35, 40, 40, 300]

Range = 300 — 30 = 270

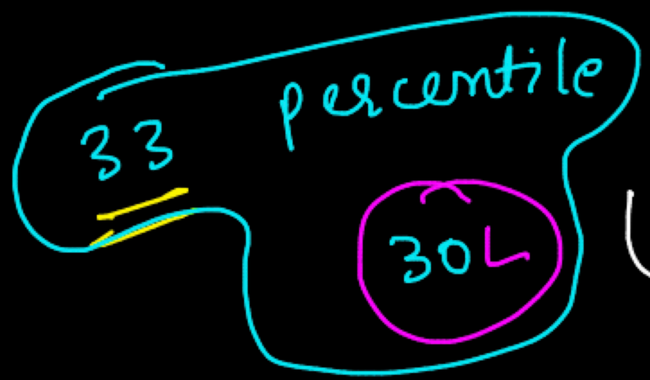
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I & R

(1) Percentile : Essentially tell you what fraction of values lesser than some specific number

salary



↳ [33% percentage of people

salary <= 30L]

IQR ⇒

case study

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30 percentile height : 150 cm

ODI

u5 / u6

Sehwag

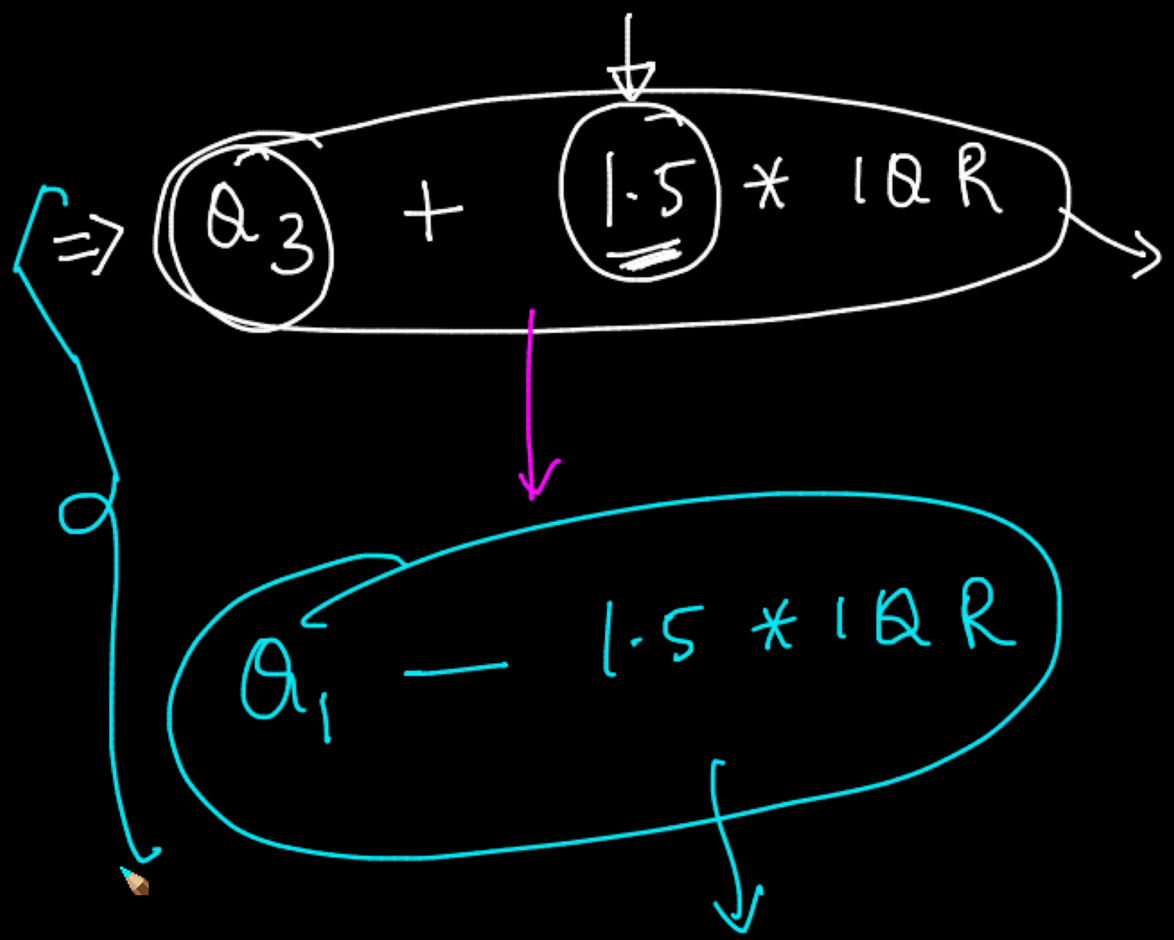
Dravid

statis

⇒ who is more Consistent ?

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=> outlier

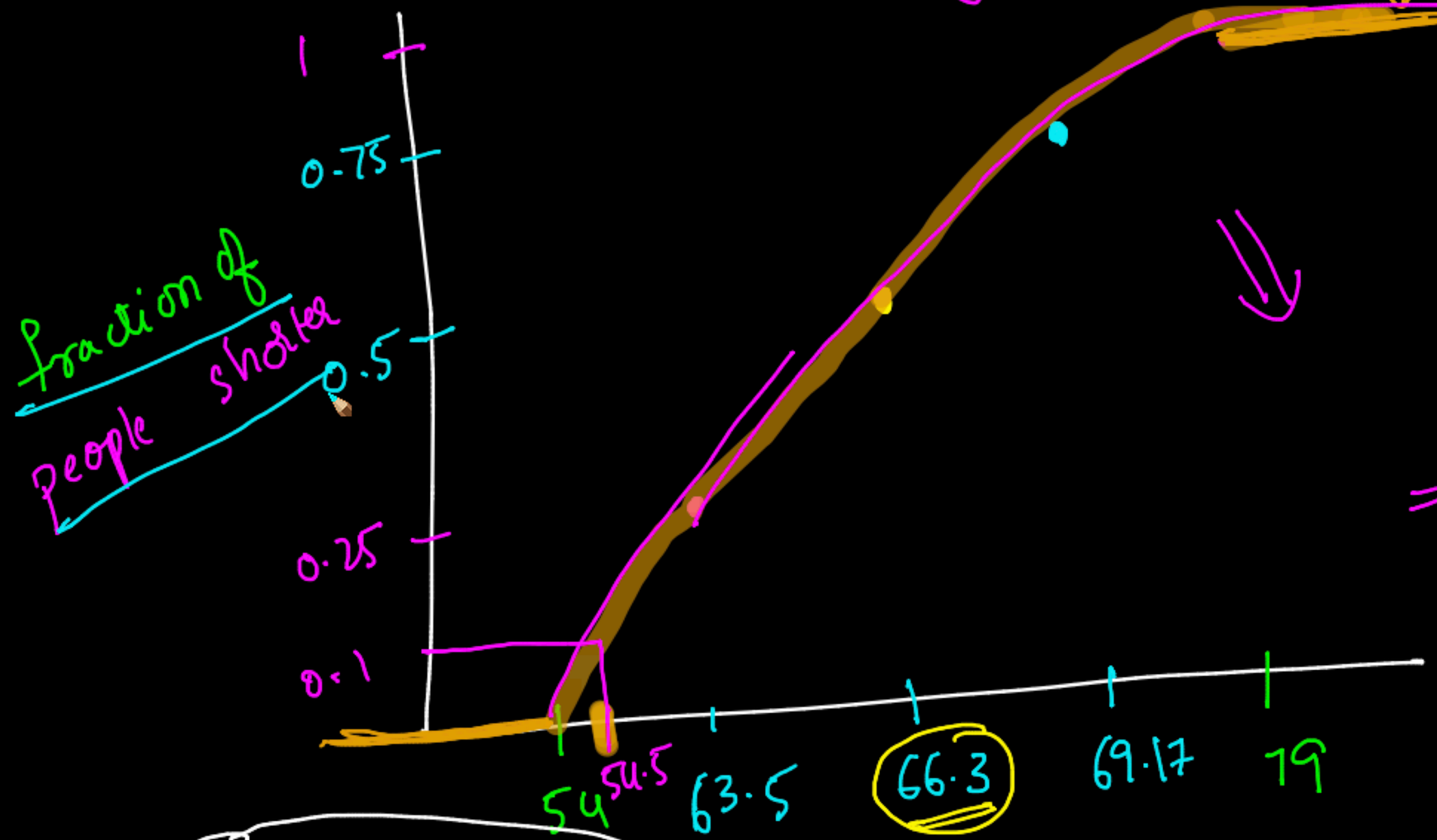
upper whisker

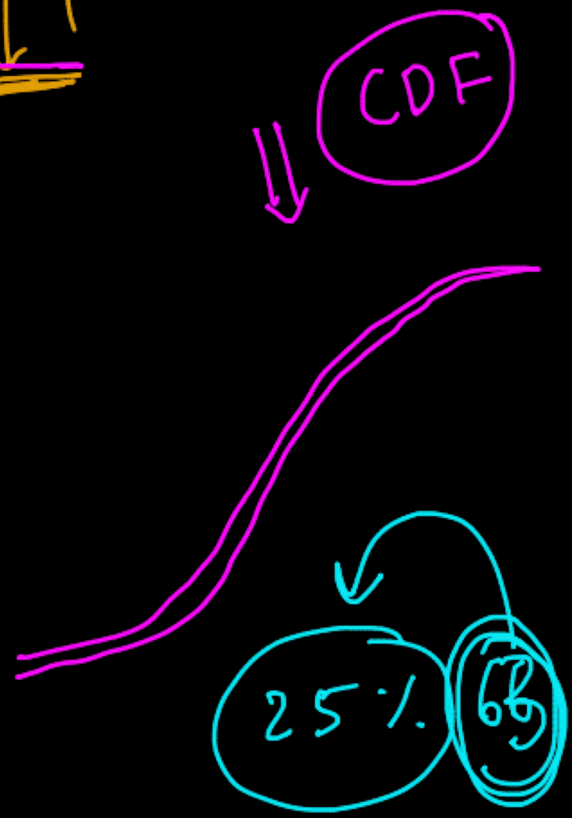
lower whisker

-ve

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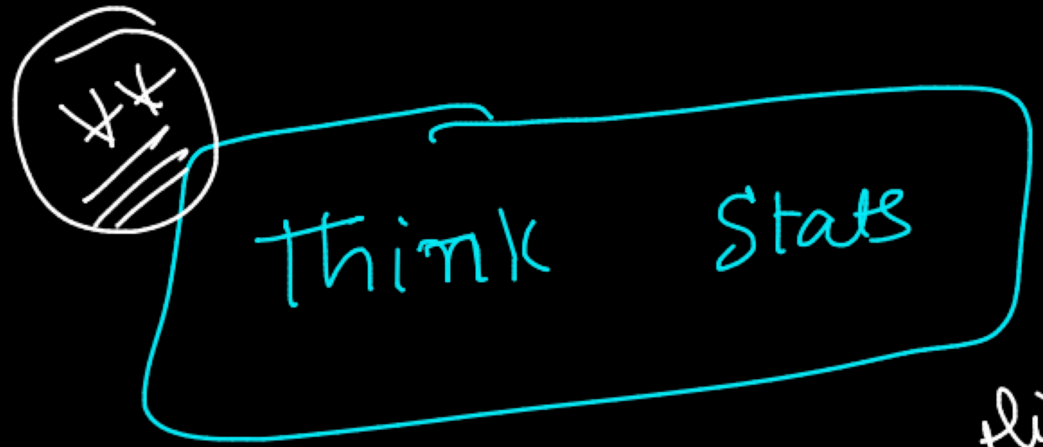




x_values = (54, 54.1, 54.2 ---- 79)

y_values = (0 ↓ ↓ 1)

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outlier

upper $Q_3 + 1.5 * IQR$

outlier $Q_1 - 1.5 * IQR$

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90, 15, 90, 95, 15, 30, 35

15, 90

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